Blessed G/Michael Catholic Secondary School

2nd Semester Biology test one for Grade 12th 24/25

March, 2018 E.C Time Allowed -35 minutes

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I. Choose the best answer among the given alternatives.

B 1. Which part of the brain regulates body temperature and Homeostasis?

A. Thalamus B. hypothalamus C. pons D. cerebellum

A 2. What is the primary function of myelin sheath?

A. To increase the speed of nerve impulses ✓ C. To protect the cell body ✓

B. To nourish the neuron. D. To produce neuro transmitters ✓

B 3. A bundle of axons in the peripheral nervous system is called

A. Tract ✓ B. nerve ✓ C. Ganglion ✓ D. synapse ✓

A 4. The cells that support, nourish and protect neurons are known as

A. Neuroglia ✓ B. synapses C. Ganglion D. Cell body

A 5. During an action potential which ion is primarily responsible for the rapid depolarization phase

A. Na^+ B. Ca^{2+} C. K^+ D. Cl^-

C 6. Which structure ensures that the left and right cerebral hemi spheres can communicate?

A. Hypothalamus B. Thalamus C. Corpuscallosum D. Pons

D 7. which of the following is a characteristic of the parasympathetic nervous system?

A. Constricts pupils ✓ B. stimulates digestion ✓ C. Contracts bladder ✓ D. all

A 8. What occurs immediately when the membrane potential reaches the threshold?

A. Voltage gated sodium channels opens rapidly ✓ C. The sodium potassium pump stops

B. Voltage gated potassium channels channels close D. potassium channels open

C 9. which event causes the repolarization phase of the action potential?

A. Influx of sodium ions C. Efflux of potassium ions

B. Influx of calcium ions D. closure of all ion channels

B 10. The all – or- none- property of an action potential means

A. All neurons fire at the same time

B. A stimulus must be strong enough to reach threshold to trigger a full action potential ✓

C. An axon potential only occurs at the axon hillock

D. The intensity of a stimulus determines the magnitude of the action potential

D 11. Cerebero spinal fluid can be described as all of the following except

A. functioning to cushion to the brain ✓ C. Maintains intra cerebral pressure ✓

B. filling cavities in the brain called ventricles ✓ D. formed from layers of connective tissue

C 12. Which of the following contains regulatory centers for the respiratory and circulatory systems

A. Hypothalamus B. Thalamus C. Medulla oblongata D. Cerebellum

A 13. Which of the following parts of the brain regulates hunger and thirst?

A. hypothalamus B. thalamus C. cerebellum D. cerebrum

D 14. If you were writing an essay which part of the brain would be most active?

A. occipital lobe B. parietal lobe C. temporal and frontal lobes D. Broca's area (Language)

A 15. Which part of the brain regulates the sleep wake up cycle (circadian rhythm)

A. pons B. thalamus C. hypothalamus D. cerebellum

D 16. The sympathetic division of the autonomic portion of the peripheral nervous system does all of the following except

A. relaxing bronchi in lungs ✓ C. Inhibiting bladder emptying ✓

B. Stimulating glucose release ✓ D. Stimulates digestion

D 17. Lysergic acid diethylamide starts to take effect around 30 to 90 minutes after taking it and physical effects include

A. dilated pupils. B. raised body temperature C. sleeplessness. D. All

D 18. Which of the following is an example of a substance use disorder

A. alcohol related disorder B. cannabis related disorder C. Inhalant related disorder D. All

D 19. The term psychological dependence is used when

A. It is clear that the individual has changed their life use of the drug ✓

B. Their activities are centered on the drug and its use ✓

C. leads to neglect of other important activities such as work, social and family commitments ✓

D. All

C

B 20. Which muscles contracts as a result of the knee Jerk reflex?

A. Hamstring B. Biceps femoris C. quadriceps femoris D. Triceps

B 21. Where is the reflex center for the knee jerk reflex located?

A. Cervical spinal cord B. Lumbar spinal cord C. medulla oblongata D. sacral spinal cord

C 22. The Achilles tendon reflex (ankle jerk) is primarily used to assess which nerve roots?

A. L3-L4 B. L4-L5 C. S1-S2 D. L5-S1

A 23. What is the normal physiological response when testing the Achilles tendon reflex?

A. plantar flexion of the foot or the foot moves away from your leg C. Inversion of foot

B. Dorsiflexion of the foot ✓ D. Eversion of foot

B 24. Neuro transmitters are released into the synaptic cleft through which process?

A. Endocytosis B. Exocytosis ✓ C. passive diffusion D. facilitated different

B 25. Where are neurotransmitter receptor sites located?

A. on the presynaptic terminal C. within the synaptic cleft

B. on the post synaptic neuron ✓ D. on the axon hillock

C 26. which neurotransmitter is commonly associated with inhibitory response often causing hyperpolarization?

A. Acetylcholine B. nor adrenaline C. GABA D. Dopamine